Pine Ridge Water Treatment Plant

1 Design Basis

The Pine Ridge Water Treatment Plant is sized for a design flow of 16.3 MGD (average day) and a peak flow of 42.7 MGD on the maximum day, serving a population of 59,975 persons.

Measured over five years, the 95th percentile raw water turbidity is 28.3 NTU. Finished water turbidity shall not exceed 0.23 NTU in 95% of monthly samples.

2 Treatment Process

The flash mixer provides a detention time of 45 s at design flow, with a velocity gradient of 643 1/s. Its volume is 12,445 gal. Flocculation follows in 3 stages, for a total detention time of 24.4 min at design flow.

There are 4 clarifiers, each 166 ft long and 33 ft wide, with a side water depth of 15.9 ft.

Filtration uses 11 dual-media filters of 426 ft² each. With one filter out of service, the filtration rate is 5.8 gpm/ft². A backwash pump delivers 3,228 gpm to the filters.

Table 1. Intake pumps

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)
IP-1	5,088	129.2	182
IP-2	8,658	80.2	271
IP-3	8,148	125.9	249

3 Chemical Feed

Ferric sulfate is dosed at 18.5 mg/L on the average day and up to 71.2 mg/L on the maximum day. Chlorine is fed at 3.7 mg/L ahead of the clearwell.

4 Disinfection Options

One option was considered for primary disinfection.

Table 2. Disinfection options

	Chlorine contact tank
Capital cost ($M)	2.68
Annual energy use (MWh/yr)	n/a
Contact time at peak flow (min)	83